Springer–Borel images in classical type

From Bruhat geometry to equivariant K-theory

An exposition of the published maps, integral refinements, and a research program

Abstract

Springer morphisms turn a representation of a reductive group into a polynomial subring of its Levi representation rings. The Borel characteristic map then sends these polynomial characters to flag cohomology. We explain the geometry and algebra behind this construction, including Bruhat cells, Schubert bases, Weyl invariants, the meaning of the classical parameters, and the distinction between normalized and literal integral sources. We review the published results of Kumar, Kumar–Rogers, and Kumar–Xie, and the subsequent internally audited integral-image deductions. We then introduce equivariant cohomology, ordinary and equivariant K-theory, and the precise image problems these theories suggest. The immediate emphasis is on uniform results in the classical families A, B, C, D.

Reading guide and status. Sections 1–5 develop the background and published construction. Section 6 explains the integral refinements; Sections 7–9 develop the new targets and the available comparison results. Section 10 states the remaining program. A statement labeled a *repository deduction* means that a proof is written in the cited companion notes and has been checked internally. It does not mean external peer review or an established claim of originality. Proposed extensions are stated as problems, not as completed theorems.

1 Groups, weights, and the meaning of a parabolic

Let G be a connected complex reductive group, $T \subset B \subset G$ a maximal torus and a Borel subgroup, and $P \supset B$ a standard parabolic. Let $L \supset T$ be its Levi subgroup. We write

$$X = X(T) = \mathrm{Hom}(T, \mathbb{C}^\times), \qquad W = N_G(T)/T, \qquad W_L = N_L(T)/T.$$

A weight $\lambda \in X$ is a multiplicative character of T. Its differential is a linear function on $\mathfrak{t} = \mathrm{Lie}(T)$. Thus the same weight gives both a function e^λ on the torus and a linear function λ on its Lie algebra. Keeping these two functions distinct is essential. The roots describe the conjugation action on $\mathfrak{g}$; the simple roots α_i determine the simple reflections s_i. The fundamental weights ω_j satisfy $\langle \omega_j, \alpha_i^\vee \rangle = \delta_{ij}$.

For most integral comparisons we take G simple and simply connected. Then X is the full weight lattice. The root lattice $Q_{\mathrm{rt}} = \sum_i \mathbb{Z}\alpha_i$ is a finite-index sublattice of X. The Lie algebra and Weyl group do not determine this character lattice: Spin_N and SO_N, for example, have different character lattices. Their complex flag varieties can agree while their specified representation sources and equivariant coefficient rings differ.

A standard parabolic is specified by its *retained* simple nodes, which generate W_L. The omitted nodes specify the steps of a partial flag. One omitted node means a maximal parabolic, several omitted nodes mean a more general partial flag, all nodes omitted give G/B, and none omitted give a point. The word parabolic therefore does not restrict the discussion to Grassmannians.

In type A_{m-1}, take $G = \mathrm{SL}_m$. The quotient for steps $0 < r_1 < \cdots < r_s < m$ parametrizes chains

$$U_{r_1} \subset \cdots \subset U_{r_s} \subset \mathbb{C}^m, \qquad \dim U_{r_j} = r_j.$$

The Levi block sizes are $r_1, r_2 - r_1, \ldots, m - r_s$, and its Weyl group permutes coordinates within each block. For one step r, the variety is $\mathrm{Gr}(r, m)$, of complex dimension $r(m-r)$.

The notation in Parts II and III instead emphasizes the size of a remaining classical factor. Put $a = n - k$. The parameter dictionary is

type	published group	G/P_{n-k}	Levi factor
C_n	$\mathrm{Sp}(2n)$	$\mathrm{IG}(n-k, 2n)$	$\mathrm{GL}(n-k) \times \mathrm{Sp}(2k)$
B_n	$\mathrm{SO}(2n+1)$	$\mathrm{OG}(n-k, 2n+1)$	$\mathrm{GL}(n-k) \times \mathrm{SO}(2k+1)$
D_n	$\mathrm{SO}(2n)$	$\mathrm{OG}(n-k, 2n)$	$\mathrm{GL}(n-k) \times \mathrm{SO}(2k)$

Here IG uses an alternating form and OG a symmetric form; the subspace must be isotropic for that form. Thus n is the rank, $n - k = a$ is the isotropic subspace dimension, and k is the residual classical rank. In the stated published ranges $1 \le k < n$ in B/C and $n \ge 4,\ 2 \le k < n$ in D. Maximal isotropic varieties require their own boundary conventions; in even orthogonal type there are two maximal isotropic components.

2 Bruhat decomposition: where the cohomology basis comes from

The Bruhat decomposition is the finite stratification

$$G/P = \coprod_{w \in W^P} C_w, \qquad C_w = BwP/P \simeq \mathbb{C}^{\ell(w)}. \tag{1}$$

Here W^P is the set of minimal representatives for the right cosets W/W_L, and $\ell(w)$ is Coxeter length, the minimum number of simple reflections in an expression for w. Minimality is equivalent to having no right descent in a retained Levi node. The closure relations among the cells define Bruhat order: $\overline{C_w} = \bigcup_{v \le w} C_v$. An algebraic filtration by these closures gives a cellular description with only even real dimensions.

There is also the opposite decomposition for the opposite Borel B^-. The opposite Schubert variety $X^w = \overline{B^- wP/P}$ has complex codimension $\ell(w)$. We use the notation of the Kumar papers:

$$\epsilon_w = [X^w] \in H^{2\ell(w)}(G/P; \mathbb{Z}), \qquad w \in W^P. \tag{2}$$

These classes form an integral basis. In particular, odd cohomology vanishes and even cohomology is torsion-free. The generating polynomial $\sum_{w \in W^P} z^{\ell(w)}$ counts the ranks of H^{2d}. Torsion in a quotient by a selected image is consequently a property of that image, not torsion in flag cohomology itself.

For $G = \mathrm{SL}_3$ and $P = B$, the six indices have lengths

ℓ	0	1	2	3
w	1	s_1, s_2	$s_1 s_2, s_2 s_1$	$s_1 s_2 s_1$

and the rank polynomial is $1 + 2z + 2z^2 + z^3$. For $\mathrm{Gr}(r, m)$, the same indices can be encoded by partitions $\lambda \subseteq (m-r)^r$: there are at most r parts, each at most $m - r$, and ϵ_λ lies in degree $2|\lambda|$. Partitions are thus finite-dimensional geometric coordinates for the Schubert basis, not parameters of the ambient group.

This basis is the bridge to exact computation. A characteristic polynomial is expanded in the ϵ_w, producing a column of integers or rationals. Products of specified source generators give a matrix whose column lattice is exactly the image we wish to understand. The use of Schubert coordinates is geometric, not an arbitrary choice of a numerical basis. Background on these decompositions and their compatibility with parabolic projections is given in Brion's lectures [5].

3 Representation rings and two kinds of variables

The representation ring $R(L) = \mathrm{Rep}(L)$ is the Grothendieck ring of finite-dimensional algebraic representations: direct sum gives addition and tensor product gives multiplication. Its elements are virtual representations, so negative coefficients are allowed. Restriction of characters to T identifies it with

$$R(T) = \mathbb{Z}[X], \qquad R(L) = \mathbb{Z}[X]^{W_L}, \qquad R(G) = \mathbb{Z}[X]^{W}. \tag{3}$$

The group ring has basis e^λ, with $e^\lambda e^\mu = e^{\lambda+\mu}$. Choosing a lattice basis makes it a Laurent polynomial ring. For a simply connected semisimple G, $R(G)$ is a polynomial ring in the fundamental irreducible characters. This does not say that every Levi ring is a polynomial ring in arbitrary torus variables.

For example, on GL_r let $t_i = e^{\epsilon_i}$. Then

$$R(\mathrm{GL}_r) = \mathbb{Z}[t_1^{\pm 1}, \dots, t_r^{\pm 1}]^{S_r} = \mathbb{Z}[e_1(t), \dots, e_r(t), e_r(t)^{-1}]. \tag{4}$$

The generator $e_i(t)$ is the character of the i-th exterior power, and $e_r(t)$ is the determinant character. The ordinary polynomial representation ring omits the determinant inverse. For SL_m the torus relation is $\prod_i t_i = 1$.

Irreducible representations are indexed by dominant highest weights. In the polynomial representation ring of the general linear group, these weights are partitions $\lambda_1 \geq \cdots \geq \lambda_r \geq 0$. Their characters are Schur polynomials $s_\lambda(t)$. The classical Grassmannian map produces the corresponding Schubert classes for partitions fitting its rectangle. This symmetric-function model is the type A construction generalized by Springer morphisms.

By contrast, put $S_{\mathbb{Q}} = \mathrm{Sym}_{\mathbb{Q}}(X \otimes \mathbb{Q})$. Its coordinates $x_i = \epsilon_i$ are additive linear functions on $\mathfrak{t}$. In type A they satisfy $\sum_i x_i = 0$. The basic global invariant presentations over $\mathbb{Q}$ are

type	$S_{\mathbb{Q}}^W$	polynomial degrees
A_{m-1}	$\mathbb{Q}[e_2(x), \dots, e_m(x)]$	$2, 3, \dots, m$
B_n, C_n	$\mathbb{Q}[e_1(x^2), \dots, e_n(x^2)]$	$2, 4, \dots, 2n$
D_n	$\mathbb{Q}[e_1(x^2), \dots, e_{n-1}(x^2), p]$	$2, 4, \dots, 2n-2, n$

(5)

Here $p = x_1 \cdots x_n$, and $p^2 = e_n(x^2)$. The B/C Weyl group allows arbitrary coordinate sign changes; type D allows only an even number. That distinction explains the Euler invariant p. Levi invariant rings combine symmetric blocks with the relevant smaller signed-permutation group.

The three expressions $e_i(t)$, $e_i(x)$, and $e_i(x^2)$ are different. They are, respectively, a torus character, a polynomial of degree i, and a polynomial of degree $2i$. Under the Borel map, polynomial degree d becomes cohomological degree $2d$. In particular, the squared-alphabet generator $e_i(x^2)$ has cohomological degree $4i$.

4 The Borel presentation and characteristic classes

Each weight determines an associated line bundle on G/B. Choose the line-bundle convention so that $\beta(\omega_i) = \epsilon_{s_i}$. The assignment of a weight to its first Chern class extends multiplicatively to the Borel characteristic map. Over $\mathbb{Q}$ its full- and partial-flag presentations are

$$\begin{aligned} H^{2*}(G/B; \mathbb{Q}) &\simeq S_{\mathbb{Q}}/(S_{\mathbb{Q},+}^W S_{\mathbb{Q}}), \\ H^{2*}(G/P; \mathbb{Q}) &\simeq S_{\mathbb{Q}}^{W_L}/(S_{\mathbb{Q},+}^W S_{\mathbb{Q}}^{W_L}). \end{aligned} \tag{6}$$

The subscript $+$ means positive polynomial degree. The relations say that positive-degree characteristic classes of global G-data vanish on the flag. Indeed, the associated bundle of a representation of the whole group is an ordinary trivial bundle.

For $\mathrm{Gr}(r,m)$, let c_i and d_j be the Chern classes of the dual tautological subbundle and dual quotient bundle, with the corresponding sign convention. An integral presentation is

$$H^*(\mathrm{Gr}(r,m);\mathbb{Z}) = \frac{\mathbb{Z}[c_1,\dots,c_r,d_1,\dots,d_{m-r}]}{((1+c_1z+\cdots+c_rz^r)(1+d_1z+\cdots+d_{m-r}z^{m-r})=1)}. \tag{7}$$

The denominator denotes all positive-degree coefficient relations. Here z is a formal bookkeeping variable; it is not an additional cohomology generator. The degree of c_i or d_i is $2i$. Eliminating one block gives the familiar complete-symmetric-function presentation and the rectangular Schur basis.

The rational presentation (6) must not be used indiscriminately over $\mathbb{Z}$. In B/D, integral Schubert classes can require factors of two relative to simple Chern-polynomial representatives. A rational invariant calculation determines a rational image, not automatically its integral lattice. Nor can one average over W_L integrally without accounting for division by $|W_L|$.

For actual calculations, the BGG operator

$$\partial_i f = \frac{f - s_i f}{\alpha_i} \tag{8}$$

lowers polynomial degree by one. The braid relations make ∂_w independent of a reduced word. For homogeneous f of degree d, its coefficient at ϵ_w, $\ell(w)=d$, is obtained by the corresponding constant divided difference, with the chosen operator-order convention. Independently, divisor multiplication can be computed by the Chevalley formula. Agreement between these routes is a useful exact check; counting reduced words as independent contributions would give the wrong answer.

5 The Springer morphism and the published maps

Let $\rho : G \to \mathrm{GL}(V)$ have faithful differential. For simple G, the trace form $B_\rho(H,K) = \mathrm{Tr}(d\rho(H)d\rho(K))$ is nondegenerate. Identify $\mathfrak{g}$ with its image in $\mathrm{End}(V)$, and project $\rho(g)$ orthogonally onto this image. The resulting regular conjugation-equivariant map is

$$\theta_\rho : G \longrightarrow \mathfrak{g}. \tag{9}$$

In the semisimple normalization used here, $\theta_\rho(1)=0$ and $d\theta_\rho|_1 = \mathrm{id}$. Its torus restriction $\Theta_\rho : T \to \mathfrak{t}$ is W-equivariant and dominant. The construction is also the trace-projection generalized Cayley map studied by Kostant–Michor [11].

Rogers's explicit weight formula makes (9) computable [4]. Write a representation weight as the column $\mu = (\mu(\alpha_1^\vee),\dots,\mu(\alpha_r^\vee))^t$, with multiplicity m_μ, and put

$$M_\rho = \sum_\mu m_\mu\, \mu\mu^t, \qquad v_\rho(t) = \sum_\mu m_\mu e^\mu(t)\mu. \tag{10}$$

If $\Theta_\rho(t) = \sum_i a_i(t)\alpha_i^\vee$, then $a(t) = M_\rho^{-1}v_\rho(t)$. The matrix entries measure the trace pairing; the vector entries are explicit Laurent polynomials. Differentiation at the identity provides a direct normalization check.

For the defining/vector representations of B/C/D,

$$\Theta_\rho^*(x_i) = \bar{t}_i = \frac{t_i - t_i^{-1}}{2}. \tag{11}$$

For the defining representation of SL_m, the corresponding formula is $x_i \mapsto t_i - \frac{1}{m}\sum_j t_j$. The common B/C/D formula does not identify their image lattices: the Weyl groups, character lattices and integral Schubert geometries are different.

Kumar defines the polynomial representation subring by

$$\mathrm{Rep}^{\mathbb{C}}_{\rho\text{-poly}}(L) = \Theta_\rho^*(\mathrm{Sym}_{\mathbb{C}}(\mathfrak{t}^*)^{W_L}) \subset R(L)\otimes\mathbb{C}, \qquad \xi_\rho^P = \beta_{\mathbb{C}}^P \circ (\Theta_\rho^*)^{-1}. \tag{12}$$

This inverse is the inverse of an injective algebra map onto its image. It is not a claim that Θ_ρ has a global inverse as a morphism of varieties. Part I proves the construction and complex surjectivity for the full Levi polynomial source [1, Definition 4 and Theorem 5].

Part II restricts to the residual symplectic or odd orthogonal factor in the displayed Levi, giving maps $\xi_{n,k}$ and $\bar{\xi}_{n,k}$. Part III gives the even orthogonal map $\xi^D_{n,k}$ [2, 3]. These restricted sources need not generate the whole cohomology of the ambient flag. In the B/D notation, set

$$e_i = e_i(\bar{h}_1^2,\dots,\bar{h}_k^2), \qquad p = \bar{h}_1\cdots\bar{h}_k \quad \text{in D}, \qquad p^2 = e_k. \tag{13}$$

The variables $\bar{h}_j = (h_j - h_j^{-1})/2$ are the tail torus coordinates, corresponding to $\bar{t}_{n-k+j}$. The published squared-generator formula, also valid in C with its corresponding quotient bundle, is

$$e_i \longmapsto c_i^2 + 2\sum_{j=1}^{i}(-1)^j c_{i+j}c_{i-j}. \tag{14}$$

The c_j are the Chern classes of the indicated quotient of the trivial defining bundle by the tautological isotropic subbundle; $c_0 = 1$, and classes outside its rank are zero. This is Part II, Proposition 11 in C and Proposition 24 in B, and Part III, Theorem 3.6 in D.

The stable theorems keep k fixed and let n tend to infinity. They use a graded inverse limit of cohomology under the standard embeddings. Part II, Theorems 16 and 29, and Part III, Theorem 3.8, prove stable injectivity of the respective maps. Finite-rank kernels remain: a finite-dimensional flag cannot support nonzero classes in arbitrarily high degree. Stable injectivity and finite integral saturation are consequently different assertions.

6 Integral refinements: what our calculations determine

6.1 The source and the saturation must both be specified

Parts II and III use useful normalized polynomial $\mathbb{Z}$-forms:

$$\mathbb{Z}[e_1,\dots,e_k] \quad \text{in B/C}, \qquad \mathbb{Z}[e_1,\dots,e_k]\oplus p\mathbb{Z}[e_1,\dots,e_k] \quad \text{in D}. \tag{15}$$

The second expression is additive, with multiplication determined by $p^2 = e_k$. These are not automatically subrings of the ordinary integral representation ring: the halves in (11) matter. For the stated SO/Sp tail forms, the literal character intersection instead uses $4^i e_i$, and in D also $2^k p$. In polynomial degree d, its image is 2^d times the normalized image. This precise rescaling is part of the source audit [12]. It must not be silently transferred to arbitrary Spin invariant sources or arbitrary representations.

More generally, for the normalized trace construction on simply connected simple G, define

$$\Lambda_\rho = \{\ell \in X\otimes\mathbb{Q} : \Theta_\rho^*\ell \in \mathbb{Z}[X]\}, \qquad A_{\rho,L} = \mathrm{Sym}_{\mathbb{Z}}(\Lambda_\rho)^{W_L}. \tag{16}$$

Our torus-source deduction identifies the full integral polynomial source with the symmetric algebra of this linear lattice, in that scope. Taking integral Levi invariants and taking a Borel image are further operations: degree-one generation of the torus source does not settle either of them. In particular, (16) records the group isogeny and representation as part of the input.

Let $L_d \subset H^{2d}(G/P;\mathbb{Z})$ be an image of any specified integral source and define

$$S_d = \mathrm{Sat}(L_d) = (L_d\otimes\mathbb{Q}) \cap H^{2d}(G/P;\mathbb{Z}). \tag{17}$$

Integer basis changes diagonalize its inclusion with positive Smith factors $s_1 \mid \cdots \mid s_r$. Thus $S_d/L_d \simeq \bigoplus_i \mathbb{Z}/s_i$, with unit factors retained. The product of the factors is the index in the selected saturation. If $r < \operatorname{rank} H^{2d}$, the full ambient quotient also has a free part. For example, the image generated by $2h$ in projective cohomology has factor 2^d in degree $2d$, although h and $2h$ generate the same rational algebra.

6.2 A type A integral baseline

For defining SL_m, the literal linear lattice is Q_{rt}. On $G/P_1 = \mathbb{P}^{m-1}$, its full Levi source is generated by the elementary symmetric functions in $x_1 - x_2, \ldots, x_1 - x_m$. Their ordinary images are $\binom{m}{j}h^j$, and our integral projective-family deduction gives

$$J_H = \bigoplus_{d=0}^{m-1} \frac{m}{\gcd(m,d)}\mathbb{Z}h^d, \qquad \gcd(m,0) = m.$$

Thus the degree-$2d$ factor is $m/\gcd(m,d)$. This concerns the literal SL source; it is not the assertion that the usual GL polynomial-representation map fails to be surjective.

6.3 Type C: primitivity from quaternionic geometry

For $a = n - k \geq 1$, the normalized tail image in $H^*(\mathrm{IG}(a,2n);\mathbb{Z})$ is saturated in every degree. Its factors in degree $4m$ are all one, indexed by partitions of m inside the k-by-a rectangle. This is a deduction from classical geometry, not an assertion that the literal rescaled source is saturated.

The explanation is the compact homogeneous fibration

$$\frac{\mathrm{Sp}(n)}{U(a)\times\mathrm{Sp}(k)} \longrightarrow \frac{\mathrm{Sp}(n)}{\mathrm{Sp}(a)\times\mathrm{Sp}(k)} = \mathrm{HGr}(k,n). \tag{18}$$

Here $\mathrm{Sp}(n)$ is compact notation, whose complexification is $\mathrm{Sp}(2n)$. The fiber is $\mathrm{Sp}(a)/U(a)$. The quaternionic base has integral cohomology $\mathbb{Z}[e_1,\ldots,e_k]/(h_{a+1},\ldots,h_n)$, with $\deg e_i = 4i$; the h_j are complete symmetric functions in the squared-root alphabet. Its generators pull back to the squared-tail generators. The integral Serre spectral sequence has free terms in even bidegrees only. It collapses, and its edge image is primitive because the other filtration quotients are free. The ingredients are classical quaternionic Schubert theory; see [6, 7]. Our selected-image formulation is recorded with its source audit in [12].

The two-step type C extension uses normalized middle Chern generators and squared-tail generators. The joint image is saturated, the sum of the two separate images is primitive, and their intersection is the constants. Compatible integral bases, rather than rational dimension counts, prove these statements. They are applications of published integral Schubert and Leray–Hirsch theory.

6.4 Type B: a complete partition formula

The audited type B deduction determines every factor for the normalized map $\bar{\xi}^{\mathbb{Z}}_{n,k}$. Put

$$\ell = \min(k, n-k), \quad A = \max(k, n-k), \quad r = \lfloor n/2 \rfloor, \quad c = \lceil n/2 \rceil.$$

For $\gamma \subseteq A^\ell$, including zero parts, define

$$b_i = \gamma_i + \ell - i, \qquad u(\gamma) = \#\{i : b_i < r\}, \qquad E_{n,k}(\gamma) = \#\{i : b_i \geq c + u(\gamma)\}. \tag{19}$$

Theorem 6.1 (Repository deduction: normalized type B)**.** *For $1 \leq k < n$, the nonzero Smith factors in cohomological degree $4m$ are*

$$\left\{2^{E_{n,k}(\gamma)} : \gamma \subseteq A^\ell,\ |\gamma| = m\right\}. \tag{20}$$

The image vanishes in degrees not divisible by four.

This counts multiplicities; it does not claim that the original Schur polynomials themselves form a Smith basis. For instance $n = 3, k = 1$ gives the factors $1, 1, 2$ in degrees $0, 4, 8$, respectively. The proof constructs primitive Schur-product prefix lattices, identifies their ranks geometrically, and uses an adapted integral basis to recover the powers of two. Published partition counts and character identities supply inputs; the additional assertion is their integral-image interpretation.

6.5 Type D: the Euler sector and integral gluing

Define $\mathcal{B}_{n,k,e}(z) = \sum_{\gamma \subseteq A^\ell,\ E_{n,k}(\gamma)=e} z^{2|\gamma|}$, and use $\mathcal{B}_{k,k,0} = 1$, all other $\mathcal{B}_{k,k,e} = 0$. The variable z^d records cohomological degree $2d$. Set $\mathcal{I}_{n,k} = 0$ for odd k. For $k = 2h$, put

$$\mathcal{I}_{n,2h}(z) = z^{2h(\lfloor (n-2h)/2 \rfloor + 1)} \begin{bmatrix} h + \lceil (n-2h)/2 \rceil - 1 \\ h \end{bmatrix}_{z^2}. \tag{21}$$

Here $\begin{bmatrix} v \\ u \end{bmatrix}_q = \sum_{\lambda \subseteq (v-u)^u} q^{|\lambda|}$ is the Gaussian polynomial.

Theorem 6.2 (Repository deduction: normalized type D)**.** *For $n \geq 4,\ 2 \leq k < n$, let $M_D(n, k; e, d)$ count the factor 2^e in degree $2d$. Then*

$$\sum_d M_D(n, k; e, d) z^d = \mathcal{B}_{n,k,e}(z) + z^k \mathcal{B}_{n-1,k,e}(z) + (\delta_{e,1} - \delta_{e,0}) \mathcal{I}_{n,k}(z). \tag{22}$$

The first two terms are the square and Euler sectors. Their rational spans are independent, but their separate integral saturations need not sum to the saturation of the combined image. This is *integral gluing*. Every gluing direction changes $(1, 1)$ into $(1, 2)$; there is no further higher-factor correction. For $n = 4, k = 2$, degree eight has factors $(1, 2, 2)$. The image has rank three in an ambient rank-four group, so the ambient cokernel also has one free direction.

These refinements answer finite-rank arithmetic questions left undetermined by complex generator formulas and stable injectivity. They do not replace the published theorems. The companion proofs, dependency audits and source-normalization qualifications are listed in [12]. Separately, the repository has complete relative root-to-weight image calculations for all proper E6 and E7 parabolics. Those compare two images, not their absolute ambient cokernels; our present priority is the classical uniform program.

7 Equivariant cohomology: retaining the torus parameters

The left action of T on G/P leads to the Borel construction

$$H_T^*(G/P; \mathbb{Z}) = H^*(ET \times_T G/P; \mathbb{Z}), \qquad S_{\mathbb{Z}} = H_T^*(\mathrm{pt}; \mathbb{Z}) = \mathrm{Sym}_{\mathbb{Z}} X. \tag{23}$$

The space ET is contractible with a free T-action, and $BT = ET/T$. The equivariant parameters are the first Chern classes of universal torus characters over BT. They have degree two. They are formal coefficient classes, not a choice of a numerical torus point. Equivariant Schubert classes form an $S_{\mathbb{Z}}$-basis indexed by W^P.

Over $\mathbb{Q}$ the presentation is

$$H_T^{2*}(G/P; \mathbb{Q}) \simeq S_{\mathbb{Q}} \otimes_{S_{\mathbb{Q}}^W} S_{\mathbb{Q}}^{W_L}. \tag{24}$$

Write u for the left coefficient variables and x for the right characteristic variables. The balancing relations are $f(x) = f(u)$ for $f \in S_{\mathbb{Q}}^W$. Thus the ordinary global-invariant relations $f(x) = 0$ are replaced by parameter-dependent relations. Setting $u = 0$ recovers (6). For integral calculations one uses an established integral characteristic lift or a geometric presentation, rather than assuming (24) without its rational coefficients.

Example 7.1 (Projective space)**.** *For the defining* SL_m *action, choose compatible signs and write*

$$H_T^*(\mathbb{P}^{m-1};\mathbb{Z}) = \mathbb{Z}[u_1,\dots,u_m,h]/\left(\sum_i u_i, \prod_i (h-u_i)\right). \tag{25}$$

The fixed points are the coordinate lines, and restriction sends h *to* u_i *at the* i*-th point. For* $m=2$*, the relation is* $h^2 = u^2$*. Forgetting equivariance gives* $h^2 = 0$*.*

The torus fixed points of a general G/P are indexed by W/W_L. Characteristic classes restrict by Weyl substitution; in compatible coordinates this is $f(w^{-1}u)$. Restrictions along an invariant curve differ by a multiple of its root label. After appropriate localization these restrictions determine the class. They provide an independent route to the same equivariant Schubert coefficients.

The equivariant extension of the original construction is

$$\xi^P_{\rho,T}(\Theta^*_\rho f) = \beta^P_T(f). \tag{26}$$

For a specified integral source its coefficient-span image is $J^H_T = S_{\mathbb{Z}}\beta^P_T(A_{\rho,L})$. Let $I = (X) \subset S_{\mathbb{Z}}$ be the parameter augmentation ideal, $H_T = H^{2*}_T(G/P;\mathbb{Z})$, and J_H the ordinary image. There is a surjection $J^H_T/IJ^H_T \to J_H$, but its injectivity is a separate question. Since H_T is free over $S_{\mathbb{Z}}$, its kernel is

$$\ker(J^H_T/IJ^H_T \longrightarrow J_H) \simeq \mathrm{Tor}_1^{S_{\mathbb{Z}}}(H_T/J^H_T,\mathbb{Z}). \tag{27}$$

Freeness of ambient equivariant cohomology does not force freeness of the quotient by the image.

Our integral $\mathrm{SL}_4/P_1 = \mathbb{P}^3$ example makes the issue concrete. The defining literal source is $\mathrm{Sym}_{\mathbb{Z}}(Q_{\mathrm{rt}})^{W_L}$. Set $a = e_2(u), b = e_3(u)$. Its elementary root-difference generators have images

$$f_1 = 4h, \qquad f_2 = 6h^2 + a, \qquad f_3 = 4h^3 + 2ah - b. \tag{28}$$

The ordinary image is $\mathbb{Z} \oplus 4\mathbb{Z}h \oplus 2\mathbb{Z}h^2 \oplus 4\mathbb{Z}h^3$. Nevertheless $J^H_T/IJ^H_T \to J_H$ first has a $\mathbb{Z}/2$ kernel in polynomial degree five. This is an internally proved and independently computed specialization defect [13]. It shows why a classical equivariant program must compute the augmentation map, not only ambient ring presentations.

8 K-theory: replacing additive classes by bundles

The ring $K^0(G/P)$ is generated by vector bundles, with the relation $[E] = [E'] + [E'']$ for an exact sequence and multiplication by tensor product. A line bundle is invertible: $[L^{-1}] = [L]^{-1}$. Unlike cohomology, ordinary K^0 is not naturally graded by codimension. It has a support, or even skeletal, filtration. For a complex flag variety its associated graded is integral Chow cohomology, identified with H^{2*}. This is the support filtration, not an assertion about the gamma filtration of a representation ring.

The characteristic homomorphism is the associated-bundle map

$$\kappa_P : R(L) \longrightarrow K^0(G/P). \tag{29}$$

For the simply connected groups in our integral setup, its representation-ring presentation is

$$K^0(G/P) \simeq \mathbb{Z} \otimes_{R(G)} R(L). \tag{30}$$

The map $R(G) \to \mathbb{Z}$ is dimension. Thus a global representation becomes an ordinary trivial bundle of its dimension. This is the multiplicative-character counterpart of the invariant relations in (6); it is not the same polynomial quotient. The characteristic and Schubert frameworks are classical [8, 10].

For projective space, take $L = [\mathcal{O}(1)]$ and $v = L - 1$. Then

$$K^0(\mathbb{P}^{m-1}) = \mathbb{Z}[v]/(v^m), \qquad \mathrm{ch}(L) = e^h = 1 + h + \frac{h^2}{2!} + \cdots . \tag{31}$$

The Chern character identifies K-theory with even cohomology rationally, with higher-degree corrections. It is not generally an integral identification in these coordinates.

The Springer-character source already supplies a canonical integral map whenever the source characters are literal:

$$\eta_\rho^P : A_{\rho,L} \longrightarrow K^0(G/P), \qquad f \longmapsto \kappa_P(\Theta_\rho^* f). \tag{32}$$

This map keeps the actual virtual character. The old map first undoes Θ_ρ^* and then applies β^P. Their relationship is a leading-filtration comparison, not the equality of $\mathrm{ch}\,\eta_\rho^P(f)$ with $\beta^P(f)$ in all degrees.

On $\mathrm{Sp}_4/P_1 = \mathbb{P}^3$, for example, the normalized coordinate x_1 has integral cohomology image h, but its rational K-class is

$$\frac{L - L^{-1}}{2} = v - \frac{v^2}{2} + \frac{v^3}{2}. \tag{33}$$

It is not integral. The literal generator $2x_1$ is integral. This is why a normalized cohomological source cannot be imported into K-theory merely because its old Schubert coefficients were integers.

Proposition 8.1 (Repository deduction: the finite filtered comparison)**.** *For the full literal source (16), with normalized trace Springer map on simply connected simple G, the maps β^P and η_ρ^P have the same integral source kernel. Their images J_H and J_K are isomorphic as rings, strictly for polynomial degree and the induced support filtration, with*

$$\mathrm{gr}\, J_K = J_H, \qquad [K^0(G/P) : J_K] = \prod_d [H^{2d}(G/P;\mathbb{Z}) : J_H^d]. \tag{34}$$

For the proof, positive global invariants pull back to virtual global characters of dimension zero, hence vanish under κ. The identity first jet of Θ_ρ makes the induced rational map triangular with identity diagonal in the support filtration. It is therefore invertible rationally; torsion-free integral targets give equality of kernels on the literal source. Homogeneous image lifts with independent leading classes establish strictness. The determinant then gives the index identity. The detailed argument and its qualifications are in [13].

The order of a finite cokernel does not determine its Smith factors. Filtration layers can join by extensions, as two $\mathbb{Z}/2$ layers can form either $(\mathbb{Z}/2)^2$ or $\mathbb{Z}/4$. For the literal defining source on $\mathrm{Sp}_4/P_1 = \mathbb{P}^3$, put $b = L - L^{-1} = 2v - v^2 + v^3$. The image has basis $1, b, b^2, b^3$. Its inclusion matrix in $1, v, v^2, v^3$ is

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & -1 & 4 & 0 \\ 0 & 1 & -4 & 8 \end{pmatrix}.$$

Its factors are $(1, 1, 8, 8)$, whereas the ordinary cohomological factors are $(1, 2, 4, 8)$. Both indices are 64. The type A defining-coordinate comparison does preserve the total ordinary Smith factors for every parabolic, but that is a representation-specific result. The completed SL3 calculations also show that equal linear lattices for two representations need not give equal embedded K-images [13].

9 Equivariant K-images and the explicit quotient map

Equivariant K-theory remembers T-actions on bundles. Its coefficient ring is $R(T) = \mathbb{Z}[X]$, rather than $\mathrm{Sym}_{\mathbb{Z}} X$, and its characteristic presentation for our simply connected setup is

$$K_T^0(G/P) \simeq R(T) \otimes_{R(G)} R(L). \tag{35}$$

The left coefficient map sends a global representation to its restricted torus character, not its dimension. Forgetting equivariance is $e^\lambda \mapsto 1$. This is distinct from setting the additive parameters of equivariant cohomology to zero, although the augmentation-filtration comparison connects them. Schubert structure sheaves give an $R(T)$-basis; Laurent Demazure operators and fixed-point restriction formulas provide established calculation methods [8, 9].

For the Springer map, define the integral image

$$J_{\rho,T}^{K,P} = R(T)\,\kappa_T^P(\Theta_\rho^* A_{\rho,L}) \subset K_T^0(G/P). \tag{36}$$

It is this embedded submodule and subring that we seek to present. There is already a useful rational source family. Put $R_{\mathbb{Q}} = \mathbb{Q}[X]$, $A = S_{\mathbb{Q}}^{W_L}$, and $B_0 = S_{\mathbb{Q}}^{W}$. Then

$$\mathcal{A}_{\rho,T}^P = R_{\mathbb{Q}} \otimes_{B_0,\, f\mapsto f(\Theta_\rho(t))} A \longrightarrow K_T^0(G/P)_{\mathbb{Q}}, \qquad 1 \otimes f \longmapsto \kappa_T^P(\Theta_\rho^* f). \tag{37}$$

The map from B_0 is essential: the basic invariant values $f_i(\Theta_\rho(t))$ are exactly the coordinates of the induced adjoint-quotient map $T/W \to \mathfrak{t}/W$. The previous quotient-coordinate computations therefore become coefficients of relations in a characteristic-image problem. At $t = 1$ these positive-degree coefficients vanish, giving the rational ordinary source quotient. A general integral freeness statement is not being inferred from (37).

9.1 A determinant theorem and its geometric meaning

For each root-length orbit a, define

$$\Delta_a(x) = \prod_{\alpha>0,\ \text{length}=a} \alpha(x), \quad \varpi_a = \tfrac{1}{2} \sum_{\alpha>0,\ \text{length}=a} \alpha, \quad \delta_a(t) = e^{\varpi_a} \prod_{\alpha>0,\ \text{length}=a} (1 - e^{-\alpha}).$$

The root quotient $\mathcal{J}_{\rho,a} = \Delta_a(\Theta_\rho(t))/\delta_a(t)$ is a regular invariant Laurent polynomial over $\mathbb{Q}$, with value one at the identity. Let $N = |W/W_L|$, and let $N_a, N_{L,a}$ count positive roots of that length in G, L. Define

$$m_{P,a} = \frac{N}{2}\left(1 - \frac{N_{L,a}}{N_a}\right). \tag{38}$$

This integer is half the number of cosets moved by a reflection of that length.

Theorem 9.1 (Repository deduction over $R_{\mathbb{Q}}$)**.** *The map (37) is injective between finite free $R_{\mathbb{Q}}$-modules of rank N. If $C_{\rho,T}^P$ is its cokernel, then*

$$\operatorname{Fitt}_0(C_{\rho,T}^P) = \left(\prod_a \mathcal{J}_{\rho,a}^{m_{P,a}}\right). \tag{39}$$

The proof compares two fixed-point matrices. A homogeneous polynomial basis evaluates as $f_j(w^{-1}\Theta_\rho(t))$; each reflection wall forces coincident row pairs. Divisibility and the total degree determine its root-product determinant. The target Schubert restriction matrix has the torus Euler root products as determinant. Their ratio is (39), up to a unit.

For a presentation $R^b \to R^a \to C \to 0$, $\operatorname{Fitt}_i(C)$ is generated by the $(a-i)$-minors. Thus Fitt_0 is the determinant ideal in the square case; higher Fitting ideals record finer degeneracy information. The rings $\mathbb{Z}[X]$ and $\operatorname{Sym}_{\mathbb{Z}} X$ are multivariable rings, so there is no general Smith normal form over them. Smith factors are appropriate after passing to a specified DVR or to a finite integer-lattice specialization.

Write $\mathcal{J}_\rho = \prod_a \mathcal{J}_{\rho,a}$. The full Springer differential determinant has torus restriction

$$D_\rho|_T = \det(d\Theta_\rho)\,\mathcal{J}_\rho^2. \tag{40}$$

Consequently (39) detects the root-direction factors of the generalized Cayley determinant. The Cartan Jacobian is a separate factor; these image calculations do not by themselves determine the full critical divisor, target branch locus, or generic degree.

9.2 Two arbitrary-rank classical families already available

For the defining SL_m-module V, the rational equivariant map (37) is an isomorphism for every P. The trace formula gives $\Theta_{V^{\otimes j}} = (\chi_V/m)^{j-1}\Theta_V$. A homogeneous basis of degrees $\ell(w)$ therefore yields the complete rational cokernel decomposition

$$C^P_{V^{\otimes j},T} \simeq \bigoplus_{w\in W^P} R_{\mathbb{Q}}/(\chi_V^{(j-1)\ell(w)}), \qquad j \geq 1. \tag{41}$$

A term with exponent zero is the zero module. This supplies every higher Fitting ideal in this family, not just the determinant. The literal integral source remains an additional arithmetic input.

There is also a complete integral source model for defining $\mathrm{Sp}_{2n}/P_1 = \mathbb{P}^{2n-1}$. Put

$$R = \mathbb{Z}[t_1^{\pm 1}, \ldots, t_n^{\pm 1}], \quad s_i = t_i + t_i^{-1}, \quad B_i = t_i - t_i^{-1}, \quad b = L - L^{-1}.$$

Then

$$K^0_T(\mathbb{P}^{2n-1}) = R[L]/\prod_i (L - t_i)(L - t_i^{-1}), \qquad J^{K,P_1}_{\rho,T} = R[b] \simeq R[z]/\prod_i (z^2 - B_i^2). \tag{42}$$

The omitted tail source generators are eliminated by the global squared-symmetric relations. The image is free on $1, b, \ldots, b^{2n-1}$, and the integral determinant is

$$\mathrm{Fitt}_0(K^0_T/J^{K,P_1}_{\rho,T}) = \left(2^n \prod_{i<j} (s_i + s_j)^2\right). \tag{43}$$

To see the factors, compare the Vandermonde matrices for L and $L - L^{-1}$ at $t_1, t_1^{-1}, \ldots, t_n, t_n^{-1}$. Each reciprocal pair contributes two; the four cross-pairs for $i < j$ contribute $(s_i + s_j)^2$. At $t_i = 1$, the determinant is $2^{n(2n-1)}$. This family's image specializes faithfully to its ordinary K-image. Its arbitrary-rank higher Fitting ideals remain a substantive next problem. Proofs and independent finite checks of (34), (39), (41)–(43) are recorded in [13].

10 A classical program with precise next goals

The common goal is to determine the embedded integral image (36), its module presentation, its Fitting ideals and its augmentation to ordinary K-theory, for the defining/vector representation in the classical families. Equivariant ordinary cohomology is a parallel comparison target. The determinant theorem is an established starting point within the companion notes, not the remaining classification.

Type A supplies a controlled baseline: construct the actual integral source image on Grassmannians and general partial flags, retaining the difference between the root and weight lattices. The rational answer alone cannot see the integral specialization defects of Section 7. Type C next asks for all higher Fitting ideals in (42), and then for isotropic Grassmannians with subspace dimension greater than one. The known quaternionic arguments explain ordinary primitivity, but do not automatically give this equivariant multiplicative classification.

For B and D, begin with isotropic lines. The targets are the odd quadric Spin_{2n+1}/P_1 of dimension $2n-1$ and the even quadric Spin_{2n}/P_1 of dimension $2n-2$. Their K-theory must retain the full Schubert or spinor-bundle data. In D the two middle Schubert classes must both be retained. The vector representations factor through SO and have faithful differential on Spin; the coefficient lattice still records which group is chosen. A polynomial projective-bundle model cannot simply be substituted for a quadric target.

These first models should lead to isotropic Grassmannians and then arbitrary partial flags. Type D maximal isotropic components must be specified separately. Comparisons involving adjoint, tensor,

exterior-power and spin representations are later tests of which features depend only on the linear source and which depend on the full Springer character.

An effective computational implementation should return the source lattice and normalization, the coefficient ring, the actual image matrix or relations, Fitting data, and the augmentation map. Verification should compare fixed-point restrictions with Schubert/Demazure calculations, and recover the appropriate old ordinary image with its stated source. Exact small-rank checks test a proposed uniform argument; they do not replace its proof.

The resulting picture extends the maps of Parts I–III in a precise way. Their polynomial characters and Borel images remain the source of the construction. The integral work asks which lattice of cohomology those characters generate. Equivariance retains the torus parameters that ordinary cohomology discards; K-theory retains the multiplicative character and extension information that the leading Chern-class calculation discards. The unfinished task is to explain these integral images uniformly across the classical root data.

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